

Drop-relationship learning for semi-supervised facial action unit recognition

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ABSTRACT

Facial action units (AUs) recognition becomes important for facial analysis and has been widely applied in psychological research. Existing work on AU relationship learning only considers the relationship between AUs simultaneously, and does not consider the complex co-adaptation between AUs. Co-adaptation is good, but not always good. Learning the AU relationship simultaneously will inhibit the robust features of network learning. And AU relationship learning often requires additional landmark information. To solve these problems, we propose a novel AU Drop-relationship learning inspired by dropout. The AU Drop-relationship learning constructs the AU relationship units through prior knowledge. We further design the relationship regularization module to constraint the relationship between AU pairs. We randomly drop AU relationship units during training to suppress co-adaptation, forcing the network to learn more robust features. In addition, considering that there are massive unlabeled web facial images in reality, manual labeling of AU requires experts and is particularly time-consuming. To address this problem, we propose a method composed of consistency regularization and pseudo-multi-labeling for semi-supervised AU recognition. The proposed method outperforms the state-of-the-art semi-supervised and supervised methods on two widely used AU datasets (BP4D and DISFA).

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1. Introduction

Facial expression is one of the gentlest and most informative expressions, it can convey people's emotional state and intentions [7,31]. As facial expression plays an increasingly important role in diagnosing mental health [26], detecting deception [10], and detecting driver fatigue [33], the real world has developed a series of facial expression system. Among these systems, the Facial Action Coding System (FACS) [8] is most popular among researchers. Action units (AUs) coded by the FACS are defined by their appearance on the face, referring to a group of facial muscle movements. All facial muscles are almost the same. AUs are based on the movement of these muscles and play a fundamental role in formulating multiple facial expressions. Therefore, the development of an AU detection system with strong generalization ability will greatly promote the analysis of complex facial expressions.

According to facial muscle distribution and FACS system [8], some AUs will often appear together, and some AUs will hardly appear together. Since the occurrence of AUs are controlled by

facial muscles, the relationship between them depends on the distribution of facial muscles. There is a strong correlation between AUs, but also a complex co-adaptation relationship. Some existing methods use this prior knowledge to improve the performance of AU recognition [1,5,9,18]. However, these methods require additional landmark information or are limited by the existing labeled dataset. These methods learn all the relationships between AUs simultaneously and do not consider the complex co-adaptation relationship between AUs. Co-adaptation originally refers to the mutual adaptation between elements in a biological system [32]. As shown in Fig. 1(a), dropout prevents co-adaptation of the feature detectors by deleting some neurons randomly to improve neural networks [13]. We compare AU relationship to neuron and suppress the complex co-adaptation relationship between AUs by randomly deleting some of the relationships between AUs.

In addition, most of the existing AU recognition methods are in a supervised fashion [5,18,17,23,29]. However, the number of existing AU label face datasets is limited, and most of them are collected in laboratories. They are relatively single in terms of pose and illumination, which poses a challenge to build a robust facial AU detection system. At the same time, there are a large number of unlabeled facial images in the real world. Labeling AUs requires

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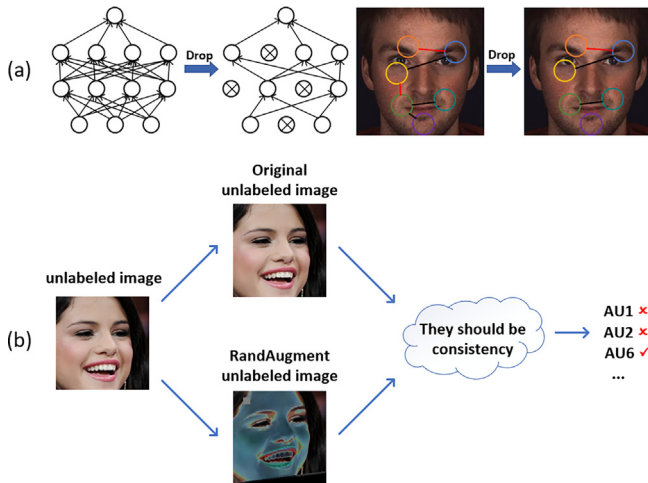


Fig. 1. (a) The illustration of the dropout and drop-relationship between AUs. We compare AU relationship to neuron and suppress the complex co-adaptation relationship between AUs by randomly deleting some of the relationships between AUs. (b) An illustration of the idea of consistency regularization. The AUs output of the RandAugment [6] unlabeled image and the Original unlabeled image through the model should be able to remain basically unchanged.

expertise and can be time-consuming. How to train a model with strong generalization ability through a small number of AU label images annotated by experts and a large number of unlabeled facial images is of great significance. Consistency regularization has become an important part of many semi-supervised learning methods [14]. The consistency regularity expresses the designer's a priori for the model that the network should be flat in the vicinity of the input data. Even if the input data changes slightly, the output of the model can remain basically unchanged. As shown in Fig. 1 (b), the AUs output of the randaugment unlabeled image and the original unlabeled image through the model should be able to remain unchanged.

In this paper, we propose an end-to-end semi-supervised network using Drop-relationship Learning module and Consistency Regularization with pseudo-multi-labeling method (called DC-Net). DC-Net aims to use AU prior knowledge and a large number of unlabeled facial images to improve AU recognition performance. In order to utilize the prior knowledge between AUs, we propose a novel AU relationship learning module by constructing AU relationship units through prior knowledge. We design a relationship regularization module to guide AU relationship learning. Inspired by dropout, we further introduced the idea of dropout in the AU relationship learning module. While learning the AU relationship, we suppress the complex co-adaptation between AUs, so that the network can learn more robust features. In addition, unlike traditional supervised learning methods, we use both AU labeled and AU unlabeled images for learning. For labeled facial images, we use traditional supervised learning to seek the supervised loss. For both the labeled and unlabeled facial images, we use the RandAugment [6] facial images and the original facial images to seek unsupervised loss through consistent regularization to guide supervised learning.

The contributions of this work are threefold: (i) we propose a novel Drop-relationship learning module, which uses prior knowledge to construct AU relationship units. We randomly drop AU relationship units during training to suppress co-adaptation. To the best of our knowledge, we are the first to consider both AU prior knowledge and complex co-adaptation relationship between AUs; (ii) we propose a new end-to-end AU semi-supervised framework, which uses consistency regularization with pseudo-multi-labeling method to extract learning signals from unlabeled facial

images; (iii) the proposed method outperforms the state-of-the-art semi-supervised and supervised methods on two widely used AU datasets (BP4D and DISFA).

2. Related Work

We investigate the existing methods relevant to our approach, including AU recognition based on relationship learning, semi-supervised and supervised AU detection, and consistency regularization.

2.1. AU Recognition based on Relationship Learning

There is a strong correlation between AUs, and the performance of AU recognition can be further improved through prior knowledge. Corneanu et al. [5] proposed a structural reasoning network based on fully connected graphs to model AU relationships. Niu et al. [22] calculated the probability of occurrence between AUs in the dataset, spread through the graph convolutional network (GCN), and established the relationship model between AUs. Li et al. [16] enhanced local features through the graph neural network (GNN) by combining the AU knowledge-graph and the ROI area of the AU. Zhao et al. [38] exploited the dependencies between AUs and facial features by introducing joint patches and multi-label learning, explored local dependencies between features through patches, and explored strong correlations across all AUs through multi-label learning. Zhang et al. [36] proposed a knowledge-driven approach to jointly learn multiple AU classifiers without any AU annotations by exploiting the prior probabilities of AUs, including both expression-independent and expression-dependent AU probabilities. Niu et al. [23] used LSTM to explore the relationship between different local regions in the feature space to improve the robustness of AU detection. However, these methods do not consider the complex co-adaptation relationship between AUs, and learning the AU relationship simultaneously will inhibit the robust features of network learning.

2.2. Supervised and Semi-supervised AU Recognition

Most of the AU research work is conducted in the form of supervision. Baltrusaitis et al. [2] developed a system for AU occurrence detection and intensity estimation by fusing geometric and appearance features. Li et al. [17] obtain the region of interest (ROI) of the AU through the facial landmark to enhance the detection performance of the AU. Zhi et al. [29] proposed a new AU detection framework through joint learning of face alignment and AU detection. Ma et al. [18] encoded prior knowledge to RCNN for AU detection. Although these supervised methods have greatly promoted the development of AU detection, the dataset with AU labels and the generalization ability is limited. Therefore, there are some methods for semi-supervised AU recognition. In [25], Peng et al. used emotion labeled images to improve the performance of AU detection by dual semi-supervised GAN. In [37], Zhao et al. pruned noisy labels by weakly supervised clustering method and used the re-annotated data to train the AU classifier. Although these methods do not require a large number of AU label images, they still need label images with other annotations, such as emotion labels.

2.3. Consistency Regularization for Semi-supervised Learning

Most state-of-art semi-supervised learning methods adopt the idea of consistent regularization. Consistency regularization indicates that the system described by the model should be smooth. Therefore, when the input data changes slightly, the output of

the model should also change slightly, and the predicted label is also approximately unchanged. The idea was initially proposed in [28,14]. Many milestones have used consistency regularization, i. e, Mean-teacher [30] and MixMatch [4]. ReMixMatch [3] used the prediction result of weakly enhanced data as guessing label, and also used a more complex loss function, so that different types of unlabeled data can be separated farther in space. VAT [20] believed that if the model can still remain smooth under the noise, then the entire network can show good consistency. For the problem of facial AU recognition, we used unlabeled data combined with consistency regularization and pseudo-multi-labeling to improve the generalization ability of the model.

3. Proposed Method

Fig. 2 shows the structure of our DC-Net for semi-supervised facial AU detection. The framework is mainly composed of three parts: the backbone model extracts features, the drop-relationship learning module performs AU relationship learning, and consistency regularization with pseudo-multi-labeling is used to extract the learning signal of unlabeled images for semi-supervised AU detection. We describe the proposed method in the following sections.

3.1. Overview of DC-Net

Feature representation is very important for building a facial AU recognition system. CNN networks such as ResNet [12] have achieved very good results in image classification, reflecting its powerful feature extraction capabilities. Therefore, we choose ResNet-34 [12] as our backbone model for feature extraction. We replaced the last 1000 -core fully connected layer of ResNet-34 with the C-core fully connected layer to match the number of AUs. For both labeled and unlabeled face images, we extract features through the same backbone model.

After we get the features generated from the backbone model, as shown in Fig. 2, the AU relationship learning module based on Dropout [13] (D-Net) is introduced to automatically explore the relationship between each AU in the feature space. Our network both considers the prior knowledge between AUs and the co-adaptation relationship between AUs. We hope that the network

can suppress the complex co-adaptation relationship between AUs while learning the AU relationship.

In order to train a model with strong generalization ability through a small number of AU labeled images annotated by experts and a large number of unlabeled facial images, we use consistency regularization with pseudo-multi-labeling method (C-Net) to Semi-supervised Facial Action Unit Recognition.

3.2. AU Relationship Units Construction

According to facial muscle distribution and FACS system, there is a strong correlation between AUs. The correlation between AUs can be further divided into positive correlation and negative correlation. Two positively correlated AUs indicate that they are likely to appear at the same time. Two negatively correlated AUs indicate that they never or rarely appear at the same time. Zhang et al. [36] summarized the positive and negative correlations of AUs, as shown in Table 1. All facial images, including facial images with and without AU labels, can apply these relationships.

To make full use of the correlation of AUs in the facial image, according to Table 1, Let $\mathcal{P}$ and $\mathcal{N}$ denote the sets of positively and negatively correlated AU pairs respectively. $\mathcal{S} = \mathcal{P} \cup \mathcal{N}$. For the relationship between M pairs of AU in $\mathcal{S}$, we construct M pairs of AU relationship units to learn the relationship between AU pairs. Each AU relationship unit corresponds to an AU pair. As shown in Fig. 3, each AU relationship unit is composed of four-layer fully connected layers to fit the correlation between AU pairs. Let $F = [X_1, X_2, \dots, X_C]$ denote the value of the feature space of C AUs obtained by the backbone model of the facial image, where $X_i, i = 1, 2, \dots, C$ is a scalar and F is a C-dimensional vector. The original AU pair X_{ij} concatenated by the i -th AU and j -th AU can be formulated as

$$X_{ij} = \text{concat}(X_i, X_j), \forall (i, j) \in \mathcal{S} \quad (1)$$

We learn the relationship between the corresponding AU pair through the AU relationship unit. New AU pair $\hat{X}_{ij}$ can be formulated as

$$\hat{X}_{ij} = \left(\prod_{k=1}^4 g(W_{k,ij}^T) \right) X_{ij}, \forall (i, j) \in \mathcal{S} \quad (2)$$

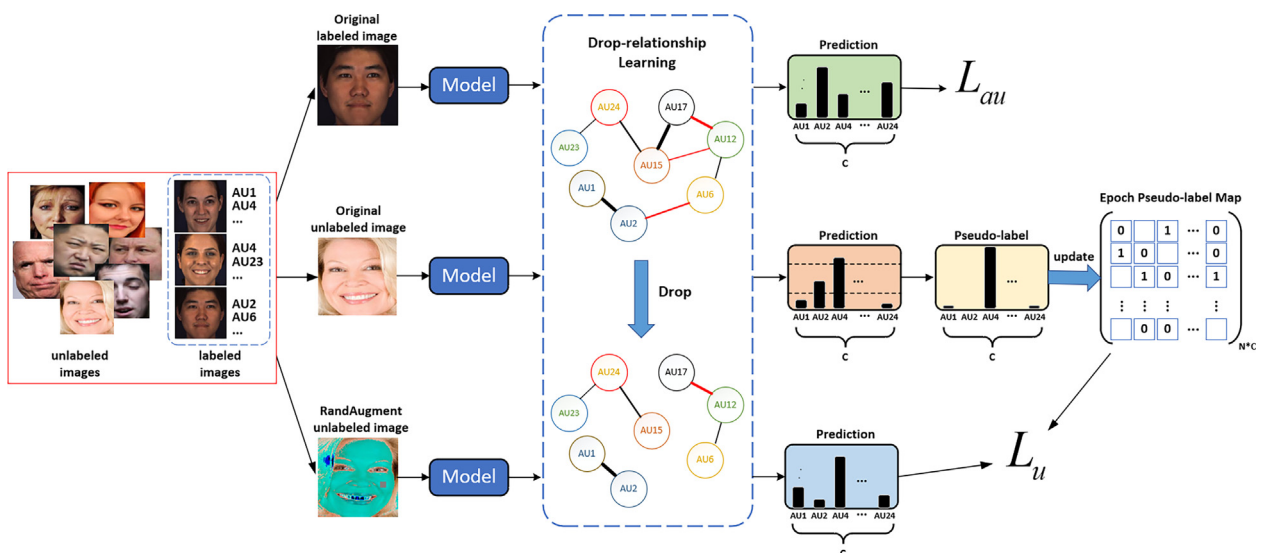


Fig. 2. An overview of our approach for semi-supervised AU recognition. Drop-relationship learning suppresses the complex co-adaptation between AUs while learning the relationship between AUs. The loss L_{au} defined for labeled facial images is used for supervised learning, and the loss L_u defined for both the labeled and unlabeled images is used for unsupervised learning through consistency regularization.

Table 1

The AU correlations according to facial muscle distribution and FACS system [8].

correlations of AU	pairs of AU
Positive correlation	(1,2), (4,7), (4,9), (7,9), (6,12), (9,17), (15,17), (15,24), (17,24), (23,24)
Negative correlation	(2,6), (2,7), (12,15), (12,17)

where $W_{k,ij} = [w_{k,ij} \ b_{k,ij}]$ is the parameters of the k -th FC layer used for learning the relationship between the i -th AU and the j -th AU pair, and $g(\cdot)$ represents the ReLU activation function added after each FC layer.

In practice, we found that removing non-linearity in the AU relationship unit can improve performance (See Fig. 6), which means that complex nonlinearities are not necessarily good. Namely,

$$\hat{X}_{ij} = \left(\prod_{k=1}^4 W_{k,ij}^T \right) X_{ij}, \forall (i,j) \in \mathcal{S} \quad (3)$$

Taking into account that after removing the nonlinearity, AU relationship unit can be replaced with a single linear layer (2×2), but it contains too few parameters to fit the relationship between AUs. Therefore, we finally use 4-layer FC layers without non-linearity in order to learn more parameters to fit the relationship between AU. We also verified this with experiments. (See Fig. 6).

For each new AU pair $\hat{X}_{ij}$ obtained in $\mathcal{S}$, decompose it into the i -th new AU value $\hat{X}_i$ and the j -th new AU value $\hat{X}_j$, i.e.,

$$(\hat{X}_i, \hat{X}_j) = \text{concat}^{-1}(\hat{X}_{ij}), \forall (i,j) \in \mathcal{S} \quad (4)$$

For repeated AU, such as AU2 in (AU1, AU2), (AU2, AU6), (AU2, AU7) will get the new value of AU2 according to Eq. 4, we take their average value as the final value. The predicted probability $\hat{p}_j$ of the final C AUs can be written as:

$$\hat{p}_j = \sigma(\hat{X}_j), \quad j = 1, 2, \dots, C \quad (5)$$

where σ is a sigmoid function, $\hat{X}_j$ is the final average value.

3.3. Relationship Regularization

In order to constraint the relationship between AU pairs, we design a relationship regularization module to regularize the weights of AU relationship units. In the relationship regularization module, we divide the output of M pairs of AU relationship units into two groups according to the positive correlation $\mathcal{P}$ and negative correlation $\mathcal{N}$. For each pair of AU relationships in the positive correlation $\mathcal{P}$, the relationship regularization guarantees two positive related AUs should be as close as possible in the prediction space, and for each pair of AU relationships in the negative correlation $\mathcal{N}$, the relationship regularization ensures that the two negative related AUs should be as far away as possible in the prediction space. Formally, we define a relationship regularization loss (RR-Loss), including positive relationship regularization loss (RRP-Loss) and negative relationship regularization (RRN-Loss), as shown below:

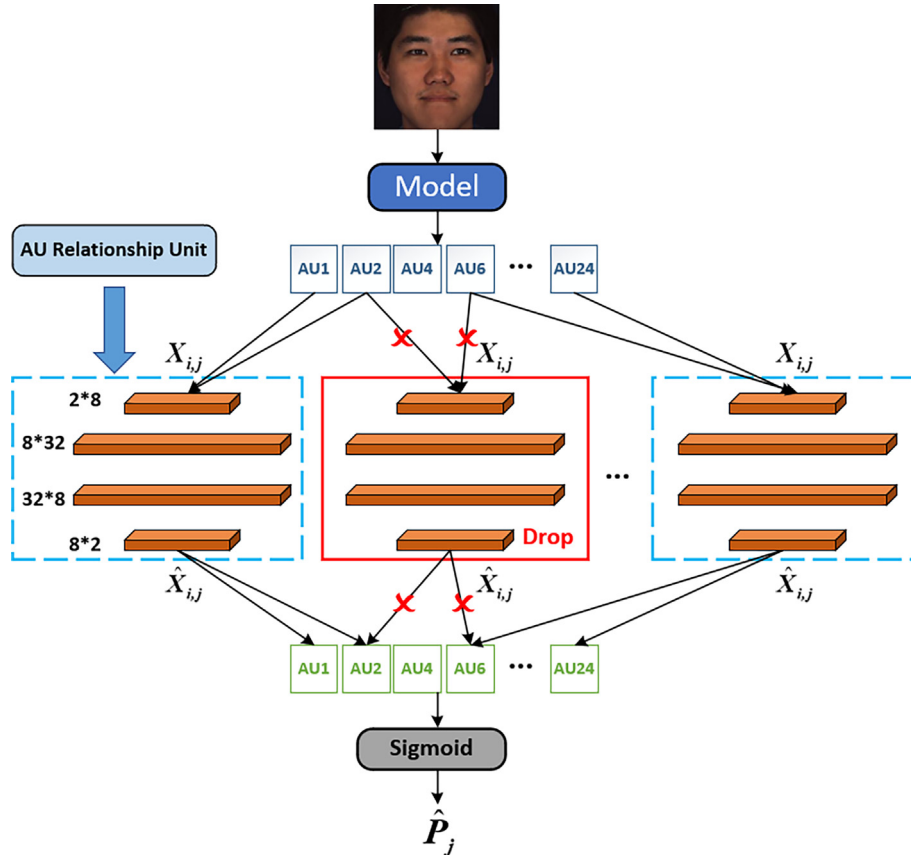


Fig. 3. The detailed structure of our Drop-relationship learning network (D-Net). AU relationship unit is composed of four-layer fully connected layers to fit the correlation between AU pairs. M pairs of AU relationship units to learn the M pairs of AU relationship. We randomly delete the AU relationship unit at a certain rate d during the training process, the red box is the AU relationship unit of the drop.

$$L_{RRP} = \sum_{\forall (i,j) \in \mathcal{P}} \max\{0, |\alpha_i - \alpha_j| - \mu\} \quad (6)$$

$$L_{RRN} = \sum_{\forall (i,j) \in \mathcal{N}} \min\{0, \mu - |\alpha_i - \alpha_j|\} \quad (7)$$

with

$$\alpha_i = \sigma\left(\frac{1}{B} \sum_{b=0}^B \hat{X}_i\right), \alpha_j = \sigma\left(\frac{1}{B} \sum_{b=0}^B \hat{X}_j\right) \quad (8)$$

where μ is a margin, B denote the batch size, α_i and α_j are the predicted mean value of the i -th AU and the mean value of the j -th AU in a batch, respectively. Finally, the total RR-Loss is $L_{RR} = L_{RRP} + L_{RRN}$.

3.4. Drop-Relationship Learning

Co-adaptation was first mentioned in biology which is the process by which multiple species, characteristics, or genes appear in pairs to adapt to each other. This interaction is only beneficial when they are together, but sometimes it can also cause them to

become interdependent. Hinton et al. [13] used dropout to prevent co-adaptation of features during neural network training.

The role of dropout is to prevent overfitting, but its essence is to reduce the interaction between feature detectors (hidden layer nodes). Detector interaction means that some detectors depend on other detectors. Co-adaptation also exists in the AU relationship. This problem also exists between AUs. According to the FACS system, we can know that the appearance of some AUs will increase the probability of other AUs, while the appearance of some AUs will reduce the probability of other AUs. Existing research does not consider the complex co-adaptation relationship between AUs. We draw the relationship between each AU into a graph in Fig. 4, in which nodes represent each AU, and edges represent the relationship between AUs, red represents positive correlation between AUs, green represents negative correlation between AUs. As can be seen from the figure, the relationship between AUs are actually very complicated, especially for learning in the network. For example, with AU4 as the starting point of the graph, through the depth-first search algorithm, it will pass through AU7, AU9, AU17, AU15, AU12, etc., and the AU nodes passing through will be connected to other AU nodes, so the relationship will become intricate.

Therefore, for ease of understanding, we simplify this process into Fig. 5. As shown in Fig. 5, assuming that A, B, C represent 3 kinds of AU, according to prior knowledge, (A, B), (B, C) are positive correlation, (A, C) is negative correlation. If we learn the relationship simultaneously, the appearance of A will increase B, and B will increase C, which is equivalent to A increase C, but A and C are negatively correlated. The appearance of A will inhibit C, and there will be relationship conflicts when learning. Therefore, if we randomly delete some AU relationships (B, C), then the network will not conflict when learning other relationships.

Inspired by dropout, we employ the idea of dropout to the relationship learning module and proposed a novel drop-relationship learning module. We compare the AU relationship unit to a neuron, and randomly delete the AU relationship unit at a certain rate d during the training process, as shown in Fig. 3, the red box is the AU relationship unit of the drop. During the test, all AU relationship units will be used.

3.5. Consistency Regularization with Pseudo-multi-labeling for Semi-supervised AU Recognition

It is very difficult to build a high-quality AU database, and most of the existing data are collected in laboratories, which brings challenges to building a robust AU detection system. Therefore, we hope to improve the generalization ability of the model through the combination of a large number of unlabeled face images in reality and the existing dataset. Therefore, we propose a method

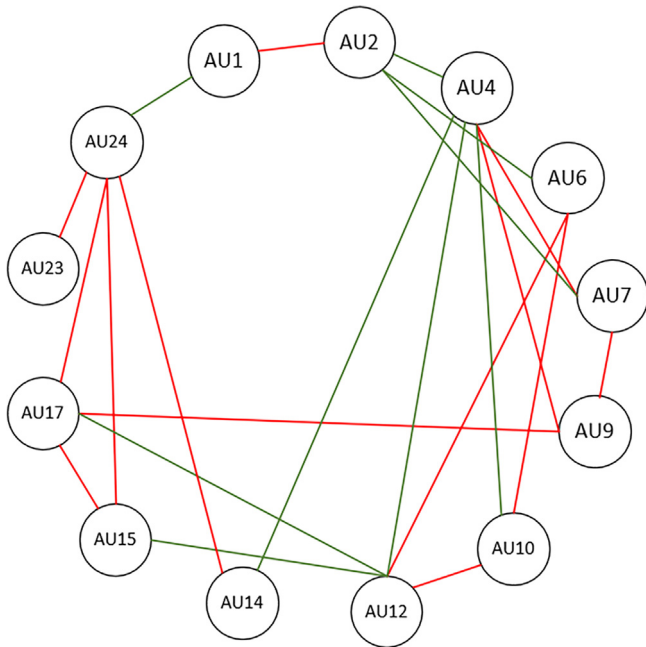


Fig. 4. The constructed AU relationship graph based on prior knowledge, where red represents positive correlation between AU and green represents negative correlation between AU.

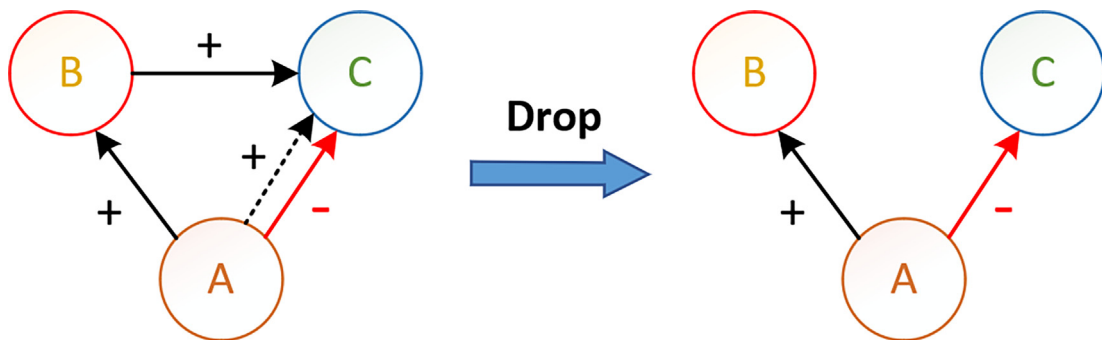


Fig. 5. Drop-relationship: the drop-out for AU level.

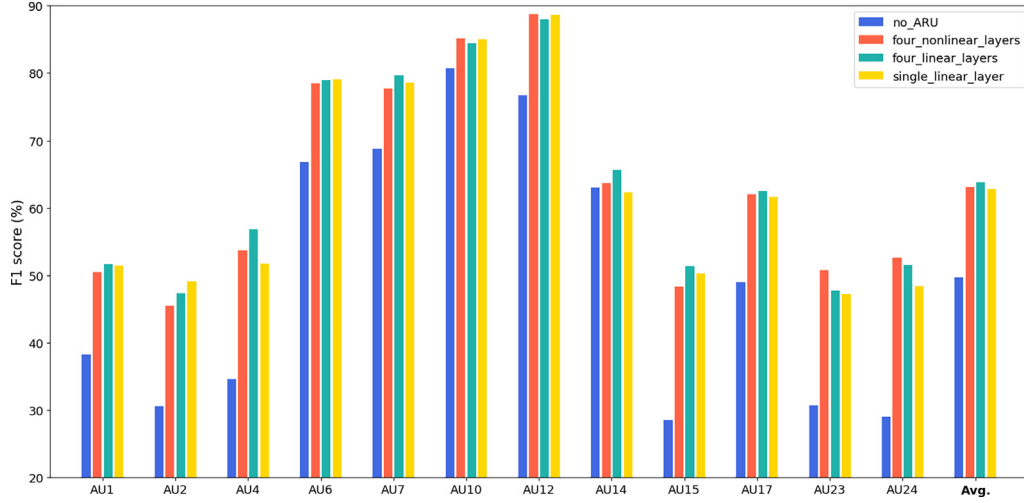


Fig. 6. Evaluation of the different structures of the AU Relationship Unit and the necessity of the AU Relationship Unit on BP4D dataset.

composed of consistency regularization and pseudo-multi-labeling for semi-supervised AU recognition.

Pseudo-multi-labeling Traditional pseudo label only retains the artificial label whose maximum category probability is higher than the predefined threshold, i.e. the $\arg \max$ output by the model [15], which is used to deal with multi-classification problems. But AU recognition is a multi-label binary classification problem. Therefore, we construct the pseudo-multi-labeling to deal with the multi-label binary classification problem. The pseudo multi-label of the j -th AU in the original unlabeled image can be formulated as

$$pl_j = \begin{cases} 1, & \text{if } \hat{p}_j - 0.5 > \tau \\ 0, & \text{if } 0.5 - \hat{p}_j > \tau \\ \text{null}, & \text{if } |\hat{p}_j - 0.5| \leq \tau \end{cases} \quad (9)$$

where *null* means no pseudo-label is given, τ means threshold hyperparameter. As shown in the orange box in Fig. 2, when the model outputs a probability to any AU higher than the threshold (upper dashed line), the prediction is converted to pseudo-label 1 (AU appears), when the model outputs a probability to any AU lower than the threshold (lower dashed line), the prediction is converted to pseudo-label 0 (AU does not appear). When the predicted probability of the model is between the given threshold (upper dashed line and lower dashed line), no pseudo-label is given.

Since we also calculate the unsupervised loss for labeled images, we construct pseudo-multi-labels for all original images. After obtaining the AU pseudo-labels of facial images, we construct an Epoch Pseudo-label Map $EPM \in R^{N \times C}$ to store the pseudo-multi-labels obtained from the previous epoch, where N represents the number of all images, C represents the number of AUs.

Consistency Regularization The consistency regularity expresses the designer's a priori for the model that the network should be flat in the vicinity of the input data. Even if the input data changes slightly, the output of the model can remain basically unchanged. The idea was initially proposed in [28,14].

For unsupervised loss, we apply it to labeled images and unlabeled images. We calculate the unsupervised loss through the EPM obtained from the original images of the previous epoch and the AU predicted probability obtained from the RandAugment [6] images of the current epoch. The unsupervised loss function L_u is formulated as

$$L_u = -\frac{1}{C} \sum_{j=1}^C w_c \left[pl_{j,t-1}^0 \log \left(\frac{\hat{p}_{j,t}^0 + 0.05}{1.05} \right) + (1 - pl_{j,t-1}^0) \log \left(\frac{1.05 - \hat{p}_{j,t}^0}{1.05} \right) \right] \quad (10)$$

where t represents the t -th epoch, O and R represent the results obtained from the Original image and RandAugment [6] image, respectively. We calculate the corresponding balance parameter w_c for each batch of neural network training through Selective Learning strategy [11]. The selective learning algorithm is used for the loss function in the deep network, which achieves the effect of data balance by weighting the samples. For example, for a batch, a label, positive samples = 70, negative samples = 30, the selective learning algorithm randomly selects 50 of the 70 positive samples as new positive samples, ignoring the other 20 positive samples. For 30 negative samples, a weight of 5/3 will be given to achieve the effect of data balance. If the positive labels are insufficient, we reversed the above process, sampling from the negatives and weighting the positives. It is worth noting that since we used the pseudo-label of the previous epoch, the unsupervised loss is calculated from the second epoch.

For the supervision loss, we calculate it through the ground truth p_j and the predicted probability $\hat{p}_j^o$ by our method. The supervised loss function L_{au} is formulated as

$$L_{au} = -\frac{1}{C} \sum_{j=1}^C w_c \left[p_j \log \left(\frac{\hat{p}_j^o + 0.05}{1.05} \right) + (1 - p_j) \log \left(\frac{1.05 - \hat{p}_j^o}{1.05} \right) \right] \quad (11)$$

The overall loss function of the proposed DC-Net can be written as

$$L_{all} = L_{au} + L_u + L_{RR} \quad (12)$$

4. Experimental Results

In this section, We conducted a series of experiments on two widely used AU datasets, i.e., BP4D [35] and DISFA [19].

4.1. Experimental Settings

4.1.1. Databases and Protocols

BP4D is a laboratory-collected, spontaneous facial expression database. It contains 41 subjects (18 males and 23 females). Each subject participated in 8 conversations and a total of 328 videos. Their facial movements are captured through 2D and 3D video. 328 videos are encoded with 12 AUs, and there are about 140,000 frames with AU labels that appear or do not appear. **DISFA** contains 27 videos of 27 subjects (15 males and 12 females). Each participant needs to watch a 4-min video and then record the AU of the face. Each video mark the intensity of 12 AU (0 to 5). The final

experiment used approximately 130,000 frames. **AffectNet** [21] queries keywords related to expression through the search engine, containing over one million of Internet-image. It provides Valence-Arousal and sentiment classification annotations, but there is no AU annotation.

Following the experiment setting of [5,17,29,23], we divide BP4D and DISFA into three folds, take two folds as training, and the other fold as a test. Repeat the experiment 3 times and get the average value. For the DISFA experiment, we fine-tuned on the best model trained on BP4D. For the DISFA dataset, we select 8 AUs for evaluation. For frames with an intensity equal to or greater than 2, we set it to positive samples and the rest to negative samples. For the unlabeled training images used in the experiment, we randomly select 100,000 images from AffectNet. We utilize MTCNN [34] to detect the face and cropped it to 224×224 on all images in the experiments.

4.1.2. Evaluation Metrics

In order to compare with the previous method [5,23], we use the F1 score to evaluate in the experiment. F1 score is an index used in statistics to measure the accuracy of a two-class model. It takes into account the accuracy and recall of the classification model at the same time, and is widely used in AU detection. We calculate the F1 scores of all AUs, and then take the average (denoted as **Avg.**).

4.1.3. Training Details

The ResNet-34 backbone of DC-Net use pre-trained ImageNet ILSVRC dataset [27] weights to initialize. We set $\tau = 0.4$, $d = 0.25$ and $\mu = 0$ as the final result according to the following ablation experiment. The initial learning rate of the SGD optimizer is set to 0.01, and the momentum is set to 0.9 to optimize DC-Net. We use a cosine annealing schedule to update learning rate. The epoch is set to 30, and the batchsize of supervised loss and unsupervised loss are both set to 100. All the experiments are performed on GeForce GTX 1080 Ti GPU using PyTorch [24].

4.2. Results

4.2.1. Comparisons with the State-of-the-art

We compare the proposed method with three state-of-the-art semi-supervised learning methods (Pseudo-Labeling [15], Mean-teacher [30] and MixMatch [4]) on the BP4D database. For Pseudo-Labeling [15], Mean-teacher [30] and MixMatch [4], we choose the same backbone network (ResNet-34) for our method for fair comparison. At the same time, we choose four state-of-the-art supervised AU recognition methods (JAA-Net [29], ROI [17], LP-Net [23], and AU R-CNN [18]) for comparison and their performance comes from [18,23].

Table 2 shows the performance of different approaches on the BP4D dataset. It can be seen that our proposed semi-supervised AU detection approach can significantly improve the performance of AU detection and is better than the state-of-the-art semi-

supervised approaches, i.e., Pseudo-Labeling [15], Mean-teacher [30] and MixMatch [4]. When comparing with ROI [17], JAA-Net [29], LP-Net [23], and AU R-CNN [18], although our approach uses additional unlabeled face images, our approach could achieve better average performance using a general ResNet-34 network without specific designs and additional features. By contrast, both ROI [17], JAA-Net [29], LP-Net [23], and AU R-CNN [18] have used additional features (facial landmarks), which helps to identify the AUs defined in small areas. In addition, the unlabeled facial images are from the wild, which are very different from BP4D in terms of illumination and posture.

We also did a corresponding experiment on DISFA dataset, and the experimental results are shown in Table 3. It can be seen that our method is again better than all the state-of-the-art semi-supervised and supervised methods. We achieved the best performance on most AUs and the best performance in the average of all AUs, which shows that our method has good generalization.

4.2.2. Specific details of the AU Relationship Unit

We explored the impact of different AU Relationship Unit structures on the final performance, including a non-linear four-layer network, a linear four-layer network, and the replacement of a four-layer linear network with a 2×2 single linear layer. In addition, we also explored the necessity of the AU Relationship Unit in DC-Net. The results are shown in Fig. 6.

Evaluation of different AU Relationship Unit structures. We first constructed a non-linear four-layer network as our AU Relationship Unit. From the results in Fig. 6, after removing the nonlinearity, the performance of the network has been improved, indicating that complex nonlinearities are not necessarily all beneficial, and after replacing the four linear layers with a single linear layer, the performance is degraded. Although they are mathematically equivalent, the dimension of the four linear layers is higher than that of the single linear layer, which is more conducive to au modeling. Second, it is difficult for AU relational units to learn specific relations between AUs only through backpropagation. Therefore, we constrain Eq. 3 by the relation regularization method RR-Loss, so that two positively correlated AUs are more likely to co-occur, while two negatively correlated AUs are less likely to co-occur. Therefore, different AU relationship units, after being constrained by the relationship regularization method RR-Loss, will finally show different effects. Therefore, we finally choose a linear four-layer network as our AU Relationship Unit.

Evaluation the necessity of the AU Relationship Unit. In order to prove that ARU is indispensable in our network, we remove the AU Relationship Unit module and directly apply L_{RR} to the output of the backbone network (denoted as no-ARU). At the same time, dropout is also directly used in the backbone network. From the results in Fig. 6, removal of the AU Relationship Unit will cause a significant decrease in network performance, which indicates that ARU is necessary in DC-Net.

Table 2

The proposed method and the state-of-the-art methods recognize the F1 score (in %) of 12 AUs on the BP4D dataset. The best is marked with brackets and bold, the second is marked with brackets.

Method	AU1	AU2	AU4	AU6	AU7	AU10	AU12	AU14	AU15	AU17	AU23	AU24	Avg.
Pseudo-Labeling [15]	43.2	44.6	50.8	77.1	76.2	83.9	86.7	61.4	42.2	61.8	41.9	43.6	59.5
Mean Teacher [30]	44.3	[46.2]	51.8	77.9	76.9	[84.7]	87.6	61.0	42.9	[62.6]	42.1	44.8	60.2
MixMatch [4]	47.1	45.3	52.8	[78.9]	76.1	[85.0]	[87.9]	62.8	44.0	60.3	44.7	45.6	60.9
Proposed	[51.6]	[47.3]	[56.8]	[79.0]	[79.7]	84.5	[88.0]	[65.6]	[51.3]	62.5	47.7	[51.5]	[63.8]
ROI [17]	36.2	31.6	43.4	77.1	73.7	85.0	87.0	62.6	45.7	58.0	38.3	37.4	56.4
DSIN [5]	[51.7]	40.4	56.0	76.1	73.5	79.9	85.4	62.7	37.3	[62.9]	38.8	41.6	58.9
JAA-Net [29]	47.2	44.0	54.9	77.5	74.6	84.0	86.9	61.9	43.6	60.3	42.7	41.9	60.0
LP-Net [23]	43.4	38.0	54.2	77.1	76.7	83.8	87.2	63.3	45.3	60.5	[48.1]	[54.2]	61.0
AU R-CNN [18]	50.2	43.7	[57.0]	[78.5]	[78.5]	82.6	87.0	[67.7]	[49.1]	62.4	[50.4]	49.3	[63.0]

Table 3

The proposed method and the state-of-the-art methods recognize the F1 score (in %) of 8 AUs on the DISFA dataset. The best is marked with brackets and bold, the second is marked with brackets.

Method	AU1	AU2	AU4	AU6	AU9	AU12	AU25	AU26	Avg.
Pseudo-Labeling [15]	38.1	40.2	60	51.7	42.9	75.1	85.6	52.4	55.6
Mean Teacher [30]	43.2	43.6	63.5	50.8	45.3	76.2	87.1	50.5	57.5
MixMatch [4]	41.9	[46.8]	61.2	[57.1]	44.8	[78]	86.2	51.2	[58.4]
Proposed	[48.7]	[51.6]	[68.2]	[55.6]	[52.3]	76.0	[92.6]	54.3	[62.4]
ROI [17]	41.5	26.4	66.4	50.7	8.5	[89.3]	88.9	15.6	48.5
DSIN [5]	42.4	39.0	68.4	28.6	46.8	70.8	90.4	42.2	53.6
JAA-Net [29]	[43.7]	46.2	56	41.4	44.7	69.6	88.3	[58.4]	56
LP-Net [23]	29.9	24.7	[72.7]	46.8	[49.6]	72.9	[93.8]	[65]	56.9
AU R-CNN [18]	32.1	25.9	59.8	55.3	39.8	67.7	77.4	52.6	51.3

4.2.3. Ablation experiment

We provide ablation experiment to evaluate the effectiveness of each part of our method. We add them step by step to the model trained only with labeled face images (denoted as Baseline). The results of the ablation experiment on BP4D are provided in Table 4. We also evaluate the influence of different hyperparameters on the experiment in Fig. 7.

Effectiveness of Consistency Regularization with Pseudo-multi-labeling. From the results in Table 4, when we add the consistency regularization with pseudo-multi-labeling (denoted as C-Net) to the baseline, we can see that the average F1 score is improved to 62.3% from 59.5%. This shows that we can learn useful information from a large number of unlabeled images in the real world, even though unlabeled images are very different from the BP4D collected in the laboratory in terms of pose and illumination.

Effectiveness of AU Relationship Units and Relationship Regularization. After adding AU Relationship Units, the experimental performance drops instead. We hope that AU Relationship Units can learn the relationship between AUs, but we believe that only constructing the corresponding AU relationship units without any constraints will not work on the network. This is also the reason why relationship regularization is added later. In order to evaluate the effectiveness of the Relationship Regularization, we conduct the experiments with and without using the regularization loss L_{RR} . If we add the regularization loss, the average F1-frame will be improved to 62.6, indicating that the constraints between AUs are useful.

Effectiveness of suppressing the co-adaptation between AUs. Finally, we added the idea of dropout to the AU relationship learning, it can be seen from Table 4 that most AUs have got better performance, and the average F1 score is improved to 63.8% from 62.6%, which proves that it is effective to suppress the complex co-adaptation between AUs. We believe that co-adaptation is good, but not always good.

Evaluation of the τ . τ is a threshold hyperparameter for constructing pseudo labels. The default τ is 0.4. We evaluate τ from

0 to 0.4 on BP4D, and show the results in Fig. 7 (left). $\tau = 0$ means that when the model predicts probability higher than 0.5 (lower than 0.5), the pseudo label is set to 1 (0), which means that all AUs of all images will be given pseudo labels. $\tau = 0.4$ means that when the predicted probability of the model is higher than 0.9 (lower than 0.1), the pseudo label is set to 1 (0), and when the predicted probability is between 0.1 and 0.9, no pseudo label is assigned. Small τ causes many incorrect pseudo-labels to be constructed which may hurt performance. Large τ constructs more correct pseudo-labels. We get the best performance in 0.4.

Evaluation of the ratio d . d is the ratio of randomly dropping AU relationship units in AU relationship learning. We study different ratios from 0 to 1.00 on BP4D. The results are shown in Fig. 7 (middle). The default ratio $d = 0.25$ obtains the best performance. $d = 0$ means that we don't drop any AU relationship unit and learn all AU relationships at the same time. This will lead to a complex co-adaptation relationship between AUs and inhibit the robust features of network learning. $d = 1$ means that we drop all AU relationship units and don't learn any AU relationships. This will result in not using the prior knowledge existing between AUs. When d is between 0–1, i.e., we not only learn the relationship between AUs, but also suppress the complex co-adaptation relationship between AUs. Therefore, it can be seen from Fig. 7 (middle) that the curve is a convex shape. When we randomly drop AU relationship units from $d = 0$ to $d = 0.25$, the average F1-frame can be improved from 62.6 to 63.8 because we suppress the complex co-adaptation relationship between AUs.

Evaluation of the μ . μ is a margin parameter to control the margin between the mean predicted probability of the i -th AU and the mean predicted probability of the j -th AU in a batch for relationship regularization. The default μ is 0. We evaluate μ from 0 to 0.2 on BP4D, and show the results in Fig. 7 (right). The default $\mu = 0$ obtains the best performance, which shows that the small margin strongly hopes that the two positively correlated AUs are closer in the predicted mean space, and the two negatively correlated AUs are farther away in the predicted mean space.

Table 4

Ablation experiments of f1 score (in %) for 12 AUs on the BP4D dataset. The best is marked with brackets and bold, the second is marked with brackets.

Method	AU1	AU2	AU4	AU6	AU7	AU10	AU12	AU14	AU15	AU17	AU23	AU24	Avg.
Baseline	48.2	42.7	51.6	76.4	75.4	83.5	87.1	60.2	41.7	61.7	43.5	42.0	59.5
Baseline + ARU w/o L_{RR}	44.2	45.2	50.3	72.2	69.3	81.1	86.2	62.2	50.9	60.3	47.1	49.2	59.8
Baseline + ARU	50.7	46.1	51.5	76.3	74.1	[86.0]	87.1	54.3	49.6	60.7	42.4	43.0	60.2
Baseline + D-Net	[56.9]	[53.0]	46.5	76.7	73.5	84.9	87.0	[65.9]	44.1	60.5	38.1	48.2	61.3
Baseline + C-Net	50.2	43.9	53.8	78.7	76.1	84.6	88.6	63.8	50.3	58.8	[47.3]	[51.1]	62.3
Baseline + C-Net + ARU w/o L_{RR}	49.5	43.2	[55.0]	78.9	[78.6]	83.7	[88.5]	63.7	50.0	[63.2]	44.0	47.6	62.2
Baseline + C-Net + ARU	50.8	42.8	54.5	[79.1]	77.5	[85.7]	[88.6]	64.4	[51.1]	61.9	46.4	48.8	[62.6]
Baseline + C-Net + D-Net	[51.6]	[47.3]	[56.8]	[79.0]	[79.7]	84.5	88.0	[65.6]	[51.3]	[62.5]	[47.7]	[51.5]	[63.8]

Baseline: ResNet34 trained only with labeled face images; **C-Net:** Consistency Regularization with Pseudo-multi-labeling; **ARU:** AU Relationship Units + L_{RR} ; **D-Net:** Drop + AU Relationship Units + L_{RR} .

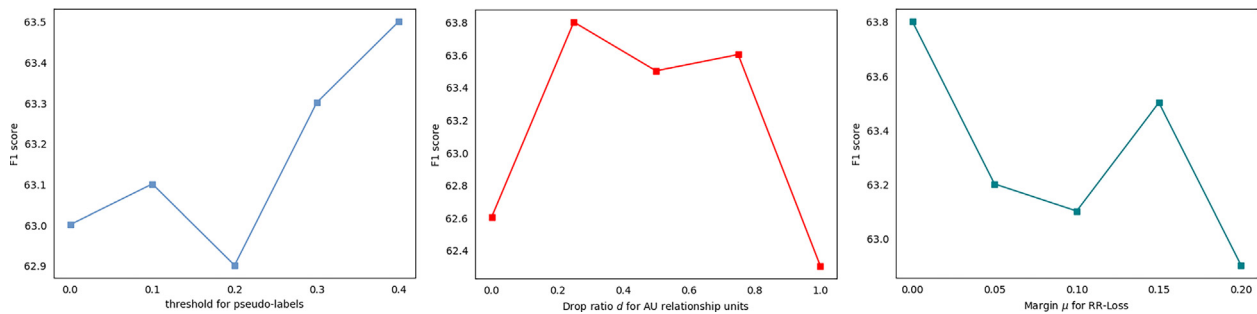


Fig. 7. Evaluation of the threshold τ , the ratio d , and the margin μ on the BP4D dataset.

5. Conclusion

In this paper, we proposed a novel Drop-relationship learning for semi-supervised facial AU recognition. Unlike existing research methods that only consider the relationship between AUs, we are learning AU relationships while suppressing the complex co-adaptation relationship between AUs. As far as we know, we are the first to consider both AU prior knowledge and Co-adaptation relationship between AUs. At the same time, there are a large number of unlabeled images in the real world. We proposed a method composed of consistency regularization and pseudo-multi-labeling, utilizing a large number of web face images without AU labels and a relatively small face dataset with AU annotations to build a more robust AU detection system. The proposed method outperforms the state-of-the-art semi-supervised and supervised methods on two widely used AU datasets (BP4D and DISFA).

CRediT authorship contribution statement

Xin Hu: Methodology, Validation, Writing - original draft, Writing - review & editing. **Ruicong Zhi:** Conceptualization, Methodology, Writing - review & editing, Supervision. **Caixia Zhou:** Data curation, Validation, Visualization.

Data availability

The data that has been used is confidential.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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